

Finitely Additive Observable Laws and the Prokhorov Extension for Query Systems

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Abstract

The observational foundations program constructs a canonical probability space from a compatible family of observable laws, taking countable additivity of those laws as a hypothesis. This paper asks when that hypothesis can itself be derived. We show that for a countable Hausdorff query system with continuous refinement maps, any finitely compatible family of finitely additive normalized valuations satisfying a *surjective projective uniform tightness* condition admits a unique extension to a σ -additive probability measure on the canonical realization space. Countable additivity is not assumed locally — it is forced globally by the combination of projective compactness and sequential refinement structure.

The result is a Prokhorov-type extension theorem in the language of query systems. It complements the main extension theorem of *Observational Foundations of Probability* (which takes σ -additive marginals as input) by identifying conditions under which those marginals need not be assumed: they emerge from a tightness condition on the compatible family as a whole. The surjective projective uniform tightness condition is the query-system analogue of Prokhorov tightness, with the surjectivity of bonding maps on compact cores playing the role that single-space compactness plays in the classical theory.

1 Introduction

The foundational question. The observational program constructs probability from observation. Its central extension theorem (*Observational Foundations of Probability*, hereafter Paper 1) shows that a compatible family of observable laws $\{\nu_Q\}$ on a query system $(\mathcal{Q}, \preceq)$ determines a unique probability measure P on the canonical realization space Ω . The construction is a representation theorem: given that each query Q carries a countably additive law ν_Q , the global P is assembled from those laws.

The question this paper asks is foundationally prior:

Must one assume countable additivity at each query level, or can it be derived from the structure of the compatible family?

This is Stopping Point 2 of the program: the assumption that each ν_Q is σ -additive is the fuel on which the extension theorem runs. The present paper identifies conditions under which that fuel can itself be derived — conditions that are properties of the *system* rather than of individual marginals.

The answer. We show that under a condition called *surjective projective uniform tightness* (SPUT), starting from only finitely additive normalized valuations $\{\ell_Q\}$ satisfying finite compatibility, one can derive a σ -additive global extension. The argument has two steps.

The first step is analytic: SPUT supplies compatible compact cores $\{K_Q\}$ whose inverse limit $K_\Omega \subseteq \Omega$ is compact and nonempty. Any decreasing sequence of cylinder sets with uniformly positive mass must meet K_Ω at each level; since K_Ω is compact, the intersection is

nonempty, contradicting the assumption that the sequence decreases to $\emptyset$. This forces the cylinder premeasure μ_0 to be σ -subadditive, hence σ -additive by finite additivity and Carathéodory extension.

The second step is geometric: the extended measure lives on Ω rather than the ambient product space. A graph-support lemma shows that for each comparable pair $Q_1 \preceq Q_2$, the joint law of $(\text{eval}_{Q_1}, \text{eval}_{Q_2})$ under P is supported on the graph of $\pi_{Q_1}^{Q_2}$. For countable $\mathcal{Q}$, a union over countably many null defect sets gives $P(\Omega) = 1$.

What SPUT is. SPUT asks that for every $\varepsilon > 0$ one can find compact sets $K_Q \subset O_Q$ at each query level such that: (i) the bonding maps restrict to surjections between compact sets, $\pi_{Q_1}^{Q_2}(K_{Q_2}) = K_{Q_1}$; and (ii) the mass outside the compact core is uniformly small, $\ell_Q(O_Q \setminus K_Q) < \varepsilon$ for every Q . Condition (i) — surjectivity of the bonding maps on the compact cores — is the critical one. It ensures the compact inverse limit $K_\Omega = \varprojlim K_Q$ is nonempty with surjective projections, which is exactly what the compactness argument requires. Without surjectivity, K_Ω can be empty even when each K_Q is nonempty.

Comparison to classical Prokhorov. In classical Prokhorov tightness [10], one controls mass concentration in a single space: for every $\varepsilon > 0$, compact $K \subset X$ with $\mu_n(X \setminus K) < \varepsilon$. The query-system analogue must control concentration *projectively*, across all query levels simultaneously, and must ensure the compact sets assemble coherently in the inverse limit. SPUT is exactly this coherent projective tightness condition, with surjectivity of bonding maps playing the role that single-space compactness plays classically [9].

Relation to Paper 1. The two extension theorems are complementary:

	Paper 1 (realizability route)	This paper (tightness route)
Input measures	σ -additive ν_Q	finitely additive ℓ_Q
Key condition	realizability of Ω	surjective projective uniform tightness
Topology required	no	Hausdorff O_Q , continuous π
Conclusion	σ -additive P on Ω	σ -additive P on Ω

Paper 1 takes σ -additivity as given and derives a global measure from realizability. This paper takes only finite additivity and derives both σ -additivity and realizability from tightness. The price is topological structure on the query system.

Structure of the paper. Section 2 fixes notation and introduces finitely additive compatible families on query systems. Section 3 defines SPUT, motivates the surjectivity condition, and gives examples. Section 4 proves σ -subadditivity on closed cylinder sequences (the intermediate theorem). Section 5 extends to general Borel cylinders using Alexandrov regularity. Section 6 establishes concentration on Ω via the graph-support lemma. Section 7 assembles the full theorem and compares it to Paper 1. Section 8 discusses what the result achieves for the foundational program, what it does not yet achieve, and the open frontier.

2 Setup

Definition 1 (Query system). A *query system* is a tuple $(\mathcal{Q}, \preceq, \{O_Q\}, \{\pi_{Q_1}^{Q_2}\})$ where:

- $(\mathcal{Q}, \preceq)$ is a countable directed preorder (the *query index set*);
- each $Q \in \mathcal{Q}$ is assigned a measurable outcome space $(O_Q, \mathcal{B}(O_Q))$;

- for each $Q_1 \preceq Q_2$, there is a measurable map $\pi_{Q_1}^{Q_2} : O_{Q_2} \rightarrow O_{Q_1}$ (the *refinement map*) satisfying $\pi_{Q_1}^{Q_1} = \text{id}$ and $\pi_{Q_1}^{Q_3} = \pi_{Q_1}^{Q_2} \circ \pi_{Q_2}^{Q_3}$ whenever $Q_1 \preceq Q_2 \preceq Q_3$.

The system is *sequentially upper-directed* if for every countable family $\{Q_n\}_{n \geq 1} \subset \mathcal{Q}$ there exists $Q_* \in \mathcal{Q}$ with $Q_* \succeq Q_n$ for all n .

Remark 1 (Topological enrichment). Throughout this paper the outcome spaces O_Q are assumed to be Hausdorff topological spaces with $\mathcal{B}(O_Q)$ the Borel σ -algebra, and the refinement maps $\pi_{Q_1}^{Q_2}$ are assumed to be continuous. These topological conditions are not required by Paper 1 and are the price paid for starting from only finitely additive marginals.

Definition 2 (Canonical realization space). The *canonical realization space* is the projective limit

$$\Omega := \varprojlim_{Q \in \mathcal{Q}} O_Q = \left\{ \omega \in \prod_{Q \in \mathcal{Q}} O_Q \mid \pi_{Q_1}^{Q_2}(\omega_{Q_2}) = \omega_{Q_1} \ \forall Q_1 \preceq Q_2 \right\},$$

equipped with evaluation maps $\text{eval}_Q : \Omega \rightarrow O_Q$, $\omega \mapsto \omega_Q$.

Since each $\pi_{Q_1}^{Q_2}$ is continuous and each O_Q is Hausdorff, each coherence constraint set $\{\omega : \pi_{Q_1}^{Q_2}(\omega_{Q_2}) = \omega_{Q_1}\}$ is closed in the product. Since $\mathcal{Q}$ is countable, Ω is a countable intersection of closed sets, hence closed in $\prod_Q O_Q$.

The *cylinder algebra* $\mathcal{C}$ is generated by sets $\text{Cyl}(Q, A) := \text{eval}_Q^{-1}(A)$ for $Q \in \mathcal{Q}$, $A \in \mathcal{B}(O_Q)$, and the *observable σ -field* is $\sigma(\mathcal{Q}) := \sigma(\mathcal{C})$.

Definition 3 (Finitely compatible family). A family $\{\ell_Q\}_{Q \in \mathcal{Q}}$ of normalized valuations $\ell_Q : \mathcal{B}(O_Q) \rightarrow [0, 1]$ (with $\ell_Q(O_Q) = 1$) is *finitely compatible* if for every $Q_1 \preceq Q_2$,

$$(\pi_{Q_1}^{Q_2})_{\#} \ell_{Q_2} = \ell_{Q_1},$$

i.e. $\ell_{Q_1}(A) = \ell_{Q_2}((\pi_{Q_1}^{Q_2})^{-1}(A))$ for all $A \in \mathcal{B}(O_{Q_1})$. Each ℓ_Q is assumed to be finitely additive (but not necessarily σ -additive).

Remark 2 (The cylinder premeasure). Given a finitely compatible family $\{\ell_Q\}$, define the *cylinder premeasure* $\mu_0 : \mathcal{C} \rightarrow [0, 1]$ by

$$\mu_0(\text{Cyl}(Q, A)) := \ell_Q(A).$$

Finite compatibility ensures μ_0 is well-defined (independent of which query level is used to represent a given cylinder) and finitely additive on $\mathcal{C}$. The central question is when μ_0 is σ -additive, so that Carathéodory's theorem [2, 3] extends it to a probability measure on $\sigma(\mathcal{Q})$.

3 Surjective projective uniform tightness

Definition 4 (Surjective projective uniform tightness). A finitely compatible family $\{\ell_Q\}$ is *surjectively projectively uniformly tight* (SPUT) if for every $\varepsilon > 0$ there exists a system $\{K_Q\}_{Q \in \mathcal{Q}}$ of compact sets $K_Q \subset O_Q$ such that:

- (i) *Surjective bonding*: $\pi_{Q_1}^{Q_2}(K_{Q_2}) = K_{Q_1}$ for all $Q_1 \preceq Q_2$;
- (ii) *Uniform mass control*: $\ell_Q(O_Q \setminus K_Q) < \varepsilon$ for every $Q \in \mathcal{Q}$.

The two conditions play distinct roles. Condition (ii) ensures that the compact cores capture most of the mass at every query level. Condition (i) ensures the compact cores assemble coherently across levels, so their inverse limit is nonempty.

Remark 3 (Why surjectivity is necessary). If condition (i) required only $\pi_{Q_1}^{Q_2}(K_{Q_2}) \subseteq K_{Q_1}$ (containment rather than equality), the inverse limit $K_\Omega := \Omega \cap \prod_Q K_Q$ could be empty even when each K_Q is nonempty. The classical example: let $O_Q = [0, 1]$, $K_Q = [0, 1 - 1/Q]$, with refinement maps the identity. Then each K_Q is nonempty and the bonding maps satisfy containment, but $\bigcap_Q K_Q = \emptyset$. Surjectivity — $\pi(K_{Q_2}) = K_{Q_1}$ exactly — rules out such “shrinking core” phenomena and guarantees $K_\Omega \neq \emptyset$ with surjective projections (Proposition 2).

Remark 4 (Comparison to classical Prokhorov tightness). Classical tightness for a family of measures $\{\mu_n\}$ on a single metric space X asks: for every $\varepsilon > 0$, a compact $K \subset X$ with $\mu_n(K) \geq 1 - \varepsilon$ for all n . SPUT is the projective analogue: compact sets must be chosen at every query level simultaneously, and they must assemble into a compact subset of Ω via the surjectivity condition. The surjectivity of bonding maps is the multi-space analogue of the fact that K is a single space (automatically surjective over itself) in the classical case.

Proposition 1 (Compact outcome spaces imply SPUT). *Suppose each outcome space O_Q is compact and each refinement map $\pi_{Q_1}^{Q_2}$ is surjective. Then every finitely compatible family $\{\ell_Q\}$ of normalized valuations is SPUT.*

Proof. For any $\varepsilon > 0$, take $K_Q := O_Q$ for every $Q \in \mathcal{Q}$. Then:

- (i) $\pi_{Q_1}^{Q_2}(K_{Q_2}) = \pi_{Q_1}^{Q_2}(O_{Q_2}) = O_{Q_1} = K_{Q_1}$ for all $Q_1 \preceq Q_2$, by surjectivity of the refinement maps.
- (ii) $\ell_Q(O_Q \setminus K_Q) = \ell_Q(\emptyset) = 0 < \varepsilon$ for every Q .

Both SPUT conditions are satisfied with the bound being 0, not merely $< \varepsilon$. □

Remark 5 (What this means). When the outcome spaces are compact, the SPUT condition is not a constraint on the valuations $\{\ell_Q\}$ at all — it is a property of the query system architecture. Any finitely compatible family, however pathological, satisfies SPUT because the compact cores are simply the full spaces. The surjectivity of refinement maps then supplies condition (i) for free. No additional tightness assumption on the observer’s laws is required.

Remark 6 (Surjective maps without compact spaces). When the outcome spaces are not compact, surjectivity of the refinement maps alone does not imply SPUT. But it does simplify the construction: given classical tightness of each ℓ_Q (compact $K_Q \subset O_Q$ with $\ell_Q(K_Q) \geq 1 - \varepsilon$), one can build compatible compact cores by setting $K'_Q := \pi_Q^{Q*}(K_{Q*})$ for a sufficiently fine Q_* . Surjectivity ensures $\pi_{Q_1}^{Q_2}(K'_{Q_2}) = K'_{Q_1}$, and the mass outside the core is controlled by the tightness of ℓ_{Q_*} . Thus the delay query systems of *Observational Probability from Delay Queries* (where refinement maps are surjective coordinate projections) satisfy SPUT whenever the marginals are individually tight.

Definition 5 (Compact core). Given a SPUT system $\{K_Q\}$, the *compact core* of Ω is

$$K_\Omega := \Omega \cap \prod_{Q \in \mathcal{Q}} K_Q = \{\omega \in \Omega \mid \omega_Q \in K_Q \ \forall Q \in \mathcal{Q}\}.$$

Proposition 2 (Properties of the compact core). *Under SPUT:*

- (i) K_Ω is compact.
- (ii) K_Ω is nonempty, and $\text{eval}_Q(K_\Omega) = K_Q$ for every $Q \in \mathcal{Q}$.

Proof. (i) The product $\prod_Q K_Q$ is compact by Tychonoff’s theorem [4]. Since Ω is closed in $\prod_Q O_Q$ (as each coherence constraint is a closed condition), $K_\Omega = \Omega \cap \prod_Q K_Q$ is a closed subset of a compact space, hence compact.

(ii) The system $(K_Q, \pi_{Q_1}^{Q_2}|_{K_{Q_2}})$ is an inverse system of nonempty compact Hausdorff spaces. By condition (i) of SPUT, the bonding maps $\pi_{Q_1}^{Q_2}|_{K_{Q_2}} : K_{Q_2} \rightarrow K_{Q_1}$ are surjective. By the *inverse limit theorem* [4, 6] for compact Hausdorff spaces with surjective bonding maps over a directed index set, $K_\Omega = \varprojlim K_Q$ is nonempty and each projection $\text{eval}_Q : K_\Omega \rightarrow K_Q$ is surjective. $\square$

Remark 7 (The inverse limit theorem). Proposition 2(ii) uses the classical result that an inverse system of nonempty compact Hausdorff spaces with surjective bonding maps has a nonempty inverse limit with surjective projections. This follows by expressing K_Ω as an intersection in the compact product $\prod_Q K_Q$, where each finite sub-intersection is nonempty by surjectivity, and applying the finite intersection property. The directed-set version of this argument is standard but is not yet available in Mathlib 4, and constitutes the main remaining gap in the Lean formalization.

4 The intermediate theorem: closed cylinders

The key analytic step is to show that the cylinder premeasure μ_0 is σ -subadditive on sequences of closed cylinders.

Theorem 1 (Closed cylinder σ -subadditivity). *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system with Hausdorff outcome spaces and continuous refinement maps. Let $\{\ell_Q\}$ be a finitely compatible SPUT family. If $C_N = \text{Cyl}(Q_N, F_N) \downarrow \emptyset$ is a decreasing sequence of cylinder sets with each $F_N \subseteq O_{Q_N}$ closed, then*

$$\mu_0(C_N) \rightarrow 0.$$

Proof. Suppose for contradiction that $\mu_0(C_N) \geq \varepsilon > 0$ for all N .

Step 1 (Compact core). Apply SPUT with parameter $\varepsilon/2$: obtain compact sets $\{K_Q\}$ with surjective bonding maps and $\ell_Q(O_Q \setminus K_Q) < \varepsilon/2$ for all Q .

Step 2 (K_Ω is compact and nonempty with surjective projections). By Proposition 2, K_Ω is compact and $\text{eval}_{Q_N}(K_\Omega) = K_{Q_N}$ for every N .

Step 3 (Each $C_N \cap K_\Omega$ is nonempty). Since $\ell_{Q_N}(F_N) = \mu_0(C_N) \geq \varepsilon$ and $\ell_{Q_N}(K_{Q_N}) \geq 1 - \varepsilon/2$, we have

$$\ell_{Q_N}(F_N \cap K_{Q_N}) \geq \varepsilon - \varepsilon/2 = \varepsilon/2 > 0.$$

In particular $F_N \cap K_{Q_N} \neq \emptyset$. Since $\text{eval}_{Q_N} : K_\Omega \rightarrow K_{Q_N}$ is surjective, there exists $\omega \in K_\Omega$ with $\text{eval}_{Q_N}(\omega) \in F_N \cap K_{Q_N}$, so $\omega \in C_N \cap K_\Omega$.

Step 4 (Each $C_N \cap K_\Omega$ is closed in K_Ω). $C_N = \text{eval}_{Q_N}^{-1}(F_N)$ where F_N is closed and eval_{Q_N} is continuous, so C_N is closed in Ω and $C_N \cap K_\Omega$ is closed in K_Ω .

Step 5 (Contradiction by compactness). The sets $\{C_N \cap K_\Omega\}$ form a decreasing sequence of nonempty closed subsets of the compact space K_Ω . By the finite intersection property,

$$\bigcap_N (C_N \cap K_\Omega) \neq \emptyset.$$

But $\bigcap_N C_N = \emptyset$ (since $C_N \downarrow \emptyset$), so $\bigcap_N (C_N \cap K_\Omega) \subseteq \bigcap_N C_N = \emptyset$. Contradiction. $\square$

5 Extension to Borel cylinders

The intermediate theorem covers closed cylinders. We now extend to all Borel cylinders. The key observation is that no new hypotheses are needed: inner regularity on each compact core is automatic.

Lemma 1 (Alexandrov regularity is free). *Let K be a compact Hausdorff space and ν a finite Borel measure on K . Then ν is inner regular: for every Borel $A \subseteq K$ and $\eta > 0$, there exists closed $F \subseteq A$ with $\nu(A \setminus F) < \eta$.*

This is Alexandrov's theorem [1, 3]; it requires only that K is compact Hausdorff, which follows from SPUT.

Theorem 2 (Borel cylinder σ -subadditivity). *Under the hypotheses of Theorem 1, if $C_N = \text{Cyl}(Q_N, A_N) \downarrow \emptyset$ is a decreasing sequence of Borel cylinder sets, then $\mu_0(C_N) \rightarrow 0$.*

Proof. Suppose $\mu_0(C_N) \geq \varepsilon > 0$ for all N . Set $\varepsilon' := \varepsilon/4$.

Apply SPUT with parameter ε' , obtaining compact cores $\{K_Q\}$ with $\ell_Q(O_Q \setminus K_Q) < \varepsilon'$ for all Q .

For each N , apply Alexandrov regularity (Lemma 1) to $\ell_{Q_N}|_{K_{Q_N}}$ with tolerance $\varepsilon'/2^N$: find closed $F_N \subseteq A_N \cap K_{Q_N}$ (closed in O_{Q_N} , since K_{Q_N} is compact in the Hausdorff space O_{Q_N}) with

$$\ell_{Q_N}(A_N \cap K_{Q_N}) - \ell_{Q_N}(F_N) < \varepsilon'/2^N.$$

Define closed cylinders $D_N := \text{Cyl}(Q_N, F_N)$. Then

$$\mu_0(D_N) = \ell_{Q_N}(F_N) \geq \ell_{Q_N}(A_N) - \varepsilon' - \varepsilon'/2^N \geq \varepsilon - 2\varepsilon' = \varepsilon/2 > 0.$$

Let $E_N := \bigcap_{k=1}^N D_k$. Since the system is sequentially upper-directed, finite intersections of closed cylinders are again closed cylinders (each pair (D_i, D_j) has a common refinement $Q_{ij} \succeq Q_i, Q_j$, giving $D_i \cap D_j = \text{Cyl}(Q_{ij}, (\pi_{Q_i}^{Q_{ij}})^{-1}(F_i) \cap (\pi_{Q_j}^{Q_{ij}})^{-1}(F_j))$, which is a closed cylinder). So each E_N is a closed cylinder.

Moreover, since C_N is decreasing and $D_k \subseteq C_k$:

$$\mu_0(E_N) \geq \mu_0(C_N) - \sum_{k=1}^N \mu_0(C_k \setminus D_k) \geq \varepsilon - \sum_{k=1}^{\infty} 2\varepsilon'/2^k = \varepsilon - 2\varepsilon'(1+1) = \varepsilon/2 > 0.$$

Since $E_N \subseteq C_N$ and $C_N \downarrow \emptyset$, we have $E_N \downarrow \emptyset$ with each $\mu_0(E_N) \geq \varepsilon/2 > 0$, contradicting Theorem 1. $\square$

Corollary 1 (σ -additivity of μ_0). *Under the above hypotheses, the cylinder premeasure μ_0 is σ -additive on $\mathcal{C}$. By Carathéodory's extension theorem, μ_0 extends uniquely to a σ -additive probability measure on $\sigma(\mathcal{Q})$ over the ambient product $\prod_Q O_Q$.*

6 Concentration on Ω

The Carathéodory extension lives on the ambient product $\prod_Q O_Q$. We must show it concentrates on the projective limit Ω .

Lemma 2 (Graph support). *Let P be the σ -additive extension of μ_0 to $\prod_Q O_Q$. For each $Q_1 \preceq Q_2$, the joint law of $(\text{eval}_{Q_1}, \text{eval}_{Q_2})$ under P is supported on the graph*

$$G_{Q_1, Q_2} := \{(a, b) \in O_{Q_1} \times O_{Q_2} \mid a = \pi_{Q_1}^{Q_2}(b)\}.$$

Proof. It suffices to verify on measurable rectangles, which generate $\mathcal{B}(O_{Q_1}) \otimes \mathcal{B}(O_{Q_2})$. For $A \in \mathcal{B}(O_{Q_1})$ and $B \in \mathcal{B}(O_{Q_2})$:

$$\begin{aligned} P(\text{eval}_{Q_1}^{-1}(A) \cap \text{eval}_{Q_2}^{-1}(B)) &= \mu_0(\text{Cyl}(Q_2, (\pi_{Q_1}^{Q_2})^{-1}(A) \cap B)) \\ &= \ell_{Q_2}((\pi_{Q_1}^{Q_2})^{-1}(A) \cap B), \end{aligned}$$

which equals $(\text{graph}_\pi)_\# \ell_{Q_2}(A \times B)$ where $\text{graph}_\pi : b \mapsto (\pi_{Q_1}^{Q_2}(b), b)$. The measure $(\text{graph}_\pi)_\# \ell_{Q_2}$ is entirely supported on G_{Q_1, Q_2} .

The graph G_{Q_1, Q_2} is Borel: it is the preimage of the diagonal $\Delta_{O_{Q_1}}$ (closed in Hausdorff $O_{Q_1} \times O_{Q_1}$) under the continuous map $(a, b) \mapsto (a, \pi_{Q_1}^{Q_2}(b))$. Hence G_{Q_1, Q_2} is closed, so the statement is well-posed. $\square$

Corollary 2. *For each $Q_1 \preceq Q_2$, defining the defect set $D_{Q_1, Q_2} := \{\omega : \omega_{Q_1} \neq \pi_{Q_1}^{Q_2}(\omega_{Q_2})\}$, we have $P(D_{Q_1, Q_2}) = 0$.*

Theorem 3 (Concentration on Ω). $P(\Omega) = 1$.

Proof. The complement decomposes as $\prod_Q O_Q \setminus \Omega = \bigcup_{Q_1 \preceq Q_2} D_{Q_1, Q_2}$. Since $\mathcal{Q}$ is countable, the preorder has countably many comparable pairs. By Corollary 2, each D_{Q_1, Q_2} is a P -null set. A countable union of null sets is null, so $P(\prod_Q O_Q \setminus \Omega) = 0$. $\square$

7 The main theorem

Theorem 4 (Prokhorov extension for query systems). *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system in which each outcome space O_Q is a Hausdorff topological space, $\mathcal{B}(O_Q)$ is its Borel σ -algebra, and each refinement map $\pi_{Q_1}^{Q_2} : O_{Q_2} \rightarrow O_{Q_1}$ is continuous. Let $\{\ell_Q\}_{Q \in \mathcal{Q}}$ be a finitely compatible family of finitely additive normalized valuations on $\{(O_Q, \mathcal{B}(O_Q))\}$.*

If $\{\ell_Q\}$ is surjectively projectively uniformly tight (Definition 4), then there exists a unique σ -additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ satisfying

$$(\text{eval}_Q)_\# P = \ell_Q$$

for every $Q \in \mathcal{Q}$. In particular, each ℓ_Q is σ -additive.

Proof. σ -additivity of μ_0 . By Theorem 2, μ_0 satisfies continuity from above at $\emptyset$ on all Borel cylinders. Since μ_0 is already finitely additive, it is σ -additive on $\mathcal{C}$.

Carathéodory extension. By Carathéodory's theorem, μ_0 extends uniquely to a σ -additive probability measure, still called P , on $\sigma(\mathcal{C})$ over $\prod_Q O_Q$.

Concentration. By Theorem 3, $P(\Omega) = 1$. The restriction $P|_\Omega$ is the required measure on $(\Omega, \sigma(\mathcal{Q}))$.

Marginal recovery. For any $Q \in \mathcal{Q}$ and $A \in \mathcal{B}(O_Q)$: $P(\text{eval}_Q^{-1}(A)) = \mu_0(\text{Cyl}(Q, A)) = \ell_Q(A)$.

Uniqueness. The observable σ -field $\sigma(\mathcal{Q})$ is generated by the cylinder algebra $\mathcal{C}$. Any two σ -additive probability measures on $(\Omega, \sigma(\mathcal{Q}))$ with the same ℓ_Q -marginals agree on $\mathcal{C}$ and hence on $\sigma(\mathcal{Q})$. $\square$

Remark 8 (Induced σ -additivity). The last sentence of Theorem 4 is the key point for the foundational program: the ℓ_Q were assumed only finitely additive, but the theorem produces a σ -additive global measure whose marginals are exactly the ℓ_Q . In particular, each $\ell_Q = (\text{eval}_Q)_\# P$ is σ -additive (as a pushforward of a σ -additive measure). Countable additivity is not an input — it is an output.

Corollary 3 (Architectural extension theorem). *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system in which each outcome space O_Q is compact Hausdorff and each refinement map $\pi_{Q_1}^{Q_2}$ is continuous and surjective. Then every finitely compatible family $\{\ell_Q\}$ of finitely additive normalized valuations admits a unique σ -additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ with $(\text{eval}_Q)_\# P = \ell_Q$ for every Q .*

Proof. By Proposition 1, compact outcome spaces and surjective refinement maps imply SPUT. Theorem 4 then applies directly. $\square$

Remark 9 (Closing Stopping Point 2). Corollary 3 is the fully structural version of Stopping Point 2. Its hypotheses are entirely architectural: they concern the topology of the outcome spaces and the surjectivity of the refinement maps, not the valuations $\{\ell_Q\}$. Any observer whose query system has compact outcome spaces and surjective refinement maps is automatically in the setting where σ -additivity is derived from the architecture, not assumed. The observer need not check whether their laws are tight, or σ -additive, or anything beyond finite compatibility.

This is the strongest form of the result: *the structure of the observational interface alone forces countable additivity of the resulting global measure.*

Remark 10 (Hypothesis audit). Every hypothesis in Theorem 4 is used, and each at exactly one point:

Hypothesis	Where used	Can it be weakened?
$\mathcal{Q}$ countable	Concentration (§6)	Countable determining subsystem suffices
Sequentially upper-directed	Borel extension (§5)	Needed for closed-cylinder intersections
Hausdorff O_Q	Ω closed; graphs Borel	Essential
Continuous π	Ω closed; cylinders closed	Essential
Surjective bonding on K_Q	$K_\Omega \neq \emptyset$	Cannot weaken to containment
Compact K_Q	Tychonoff; FIP; Alexandrov	Essential
$\ell_Q(O_Q \setminus K_Q) < \varepsilon$	Mass estimates	Essential
Finite compatibility	μ_0 well-defined	Essential

8 Discussion

What the theorem achieves. Theorem 4 addresses Stopping Point 2 of the observational program: it shows that σ -additivity need not be assumed at each query level. Instead, it is a global consequence of two structural conditions: finite compatibility (the marginals cohere under refinement) and SPUT (the mass concentrates projectively on compatible compact cores). These conditions are properties of the *system* of observable laws, not of individual marginals.

The result is a strict weakening of the input to Paper 1: Paper 1 assumes σ -additive marginals and realizability, while this paper needs only finitely additive marginals and SPUT. The conclusion is the same: a σ -additive global measure on $(\Omega, \sigma(\mathcal{Q}))$.

What SPUT is buying. SPUT is a condition about how observable mass concentrates as one moves between query levels. It says, in operational terms: the things the observer considers *likely* at fine resolutions assemble coherently into things that are likely at coarser resolutions, and this assembly is non-degenerate (surjectivity). This is a natural condition on a query system arising from a physical process: if the system concentrates mass on a compact region at every observable resolution, that region is observable at all resolutions jointly.

What it costs. SPUT requires topological structure: Hausdorff spaces and continuous refinement maps. This is the price for starting from finitely additive marginals. Paper 1 operates without this topology, at the cost of requiring σ -additivity as an input. The two theorems occupy adjacent positions in the space of hypotheses, with this paper trading topology for weaker input measures.

The deeper question. Stopping Point 2 in its strongest form asks whether σ -additivity can be forced from *purely combinatorial* conditions on the query system, without topological assumptions. Theorem 4 does not achieve this: it replaces the assumption of σ -additivity with the assumption of Hausdorff topology plus SPUT. The question is whether SPUT itself can be derived from something more primitive.

The architectural theorem and the deeper question. Corollary 3 closes the first form of Stopping Point 2: in the compact outcome space setting with surjective refinement maps, σ -additivity is a theorem of architecture. The observer contributes only finite compatibility; the system does the rest.

This raises the next question: how restrictive is the compactness assumption? For sensor systems with bounded output spaces (voltages, pressures, counts in a finite range), compactness is natural. For infinite-range observables — real-valued sensor readings, delay vectors in $\mathbb{R}^d$, unbounded counts — it cannot be assumed. The question is what structural property of the outcome spaces substitutes for compactness.

The standard Borel route: Musiał’s theorem. The existing measure-theoretic literature provides a second architectural resolution of Stopping Point 2, complementing Corollary 3 and covering the infinite-range case.

The key concept is *perfectness* in the sense of Gnedenko and Kolmogorov [3]: a probability measure μ on $(X, \mathcal{B})$ is *perfect* if for every measurable $f : X \rightarrow \mathbb{R}$, the image measure $f_*\mu$ is inner regular with respect to compact sets on $\mathbb{R}$. Perfectness is a measure-theoretic regularity condition that does not require topology on X itself.

The connection to the infinite-range case is given by two classical facts:

- (i) *Standard Borel spaces carry only perfect measures.* If O_Q is a standard Borel space — a measurable space isomorphic to a Borel subset of a Polish space — then every Borel probability measure on O_Q is perfect [3, 6]. This follows from the Borel isomorphism theorem (any uncountable standard Borel space is measurably isomorphic to $\mathbb{R}$) together with inner regularity of finite Borel measures on $\mathbb{R}$.
- (ii) *Projective systems of perfect measures have σ -additive limits with no leakage.* Musiał [8] proved that a projective system of perfect probability measures over any directed index set admits a unique σ -additive projective limit P satisfying $P(\Omega) = 1$. No topological conditions on the spaces or maps are required beyond measurability.

Combining these facts gives the following corollary, which is a consequence of [8] applied to the query system setting.

Corollary 4 (Standard Borel architectural extension). *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system whose outcome spaces O_Q are standard Borel spaces and whose refinement maps are measurable. Then every finitely compatible family $\{\ell_Q\}$ of Borel probability measures on the query system extends uniquely to a σ -additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ with $P(\Omega) = 1$.*

Proof. Each ℓ_Q is a Borel probability measure on a standard Borel space, hence perfect by fact (i). Musiał’s theorem (fact (ii)) gives a σ -additive projective limit on $\prod_Q O_Q$ with full mass on the coherent locus. The concentration argument of Section 6 applies: the graph of each measurable refinement map $\pi_{Q_1}^{Q_2} : O_{Q_2} \rightarrow O_{Q_1}$ is measurable in $O_{Q_1} \times O_{Q_2}$ when both spaces are standard Borel (the diagonal of a standard Borel space is measurable, since such spaces are T_2 and second countable after upgrading to their Polish topology [11]). Each defect set D_{Q_1, Q_2} is therefore measurable and null, and the countable union over all comparable pairs is null. $\square$

What the two architectural corollaries together achieve. Corollaries 3 and 4 together resolve Stopping Point 2 in the following two architecturally natural settings:

Setting	Key condition	Corollary
Compact outcome spaces	Surjective refinement maps	3
Standard Borel outcome spaces	Measurable refinement maps	4

In both cases the conclusion is identical: σ -additivity is not an assumption on the observer’s laws but a consequence of the query system’s architecture. The observer need only supply finite compatibility; the spaces do the rest.

The standard Borel setting covers the main computational case of the program: delay query systems with outcome spaces $O_{Q_{d,\tau}} = \mathbb{R}^d$ are standard Borel, so any compatible family of stationary process marginals extends to a σ -additive global measure without any tightness assumption.

The leakage perspective. The Yosida–Hewitt theorem [5] provides the algebraic underpinning for both results. Every finitely additive probability measure ℓ on a measurable algebra decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ -additive and ℓ_p is *purely finitely additive* — the component that assigns positive mass to decreasing sequences of sets intersecting to empty. The *leakage* of a compatible family $\{\ell_Q\}$ is

$$\lambda(\{\ell_Q\}) := 1 - P(\Omega),$$

the total mass of the purely finitely additive parts that fails to concentrate on the coherent realization space Ω .

Stopping Point 2 asks when the query system architecture forces $\lambda = 0$ for every compatible family. The compact and standard Borel cases are two settings where this holds. The open question — the remaining form of Stopping Point 2 — is whether there are weaker architectural conditions on $(\mathcal{Q}, \preceq, \{\pi_{Q_1}^{Q_2}\})$ that also force $\lambda = 0$, applicable to spaces that are neither compact nor standard Borel.

Open problems.

1. *The topology-free route.* Both architectural corollaries use some structure on the outcome spaces: topology (compact Hausdorff) in Corollary 3, or measurable-space isomorphism to a Polish subspace (standard Borel) in Corollary 4. The fully general question is whether $\lambda = 0$ can be forced from purely order-theoretic conditions on $(\mathcal{Q}, \preceq)$, with no structural assumptions on the O_Q . The Mallory–Sion theorem [7] suggests the obstruction is measurability of the coherent locus Ω in the ambient product, and that this measurability is not guaranteed without additional structure.
2. *The uncountable case.* Concentration (Theorem 3) uses countability of $\mathcal{Q}$ to take a countable union of null defect sets. For uncountable $\mathcal{Q}$, the correct generalization is a condition on determining subsystems: if $\mathcal{Q}$ admits a countable determining subsystem $\mathcal{Q}_0$ (coherence on $\mathcal{Q}_0$ forces coherence on all of $\mathcal{Q}$), the argument applies. Characterizing when such subsystems exist is the open problem.
3. *Lean formalization.* The Lean 4/Mathlib formalization of this paper has two outstanding gaps. First, the directed-set form of the inverse limit theorem (Proposition 2(ii)) is not yet in Mathlib 4. Second, the notion of a perfect measure (Definition 4) and Musiał’s theorem are absent from Mathlib; formalizing Corollary 4 would require adding both. The infrastructure for standard Borel spaces (`StandardBorelSpace` typeclass) and inner regularity (`Measure.InnerRegular`) is available, providing the foundation for future formalization.

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